3

Weak-coupling polarons: carrier-induced softening

With a sufficiently weak electron–phonon interaction a charge carrier in a semiconductor forms a weak-coupling polaron. The energies and the effective masses associated with states located near the extrema of a carrier's energy band are thereby shifted modestly. These effects can be calculated by treating the electron–phonon interaction as a small perturbation that links a carrier in a wide electronic energy band, $W \gg \hbar\,\omega$, with the harmonic vibrations of the solid's atoms. This perturbation scheme is valid provided that the electron–phonon interaction is too weak and the electronic energy band too wide for the carrier to self-trap.

3.1 Long-range electron–phonon interaction

With the long-range electron–phonon interaction of ionic and polar materials the carrier interacts primarily with long-wavelength phonons. In the absence of thermal phonons the energy of an electron at the bottom of a conduction band is then shifted by

$$\Delta E_{\mathrm{WC}}^{\mathrm{LR}} = -\frac{e^2}{2R_{\mathrm{p}}}\left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\right). \tag{3.1}$$

Here the "weak-coupling polaron radius" R_{p} is defined in terms of the carrier's bare mass m, the lattice constant a, and the associated electronic bandwidth parameter $t \equiv \hbar^2/2ma^2$ by

$$R_{\mathrm{p}} \equiv \left(\frac{\hbar}{2m\omega}\right)^{1/2} = \left(\frac{t}{\hbar\omega}\right)^{1/2} a, \tag{3.2}$$

and $W = 12t$ is the width of a nearest-neighbor tight-binding electronic band of a cubic crystal. This so-called polaron radius is not the radius of a self-trapped state. In

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fact, the weak-coupling polaron radius is independent of the electron–phonon coupling. This radius just describes a distance that a carrier can diffuse within the period of an atomic vibration.

This weak-coupling approach breaks down when the electron–phonon interaction strength and the electronic bandwidth are such as to produce a self-trapping potential well that can bind a carrier with an energy that exceeds the characteristic (optical) phonon energy (Fröhlich, 1963). The carrier can then circulate within its self-trapping potential well with a frequency, the binding energy divided by Planck's constant, that exceeds the phonon frequency.

The long-range electron–phonon coupling is often expressed in terms of the dimensionless Fröhlich coupling constant (Fröhlich, 1963):

$$\alpha \equiv \frac{e^2}{\hbar}\left(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\right)\sqrt{\frac{m}{2\hbar\omega}}. \tag{3.3}$$

As will become clear in Chapter 4, the Fröhlich coupling constant is essentially the square root of the ratio of the strong-coupling binding energy, that with the carrier being self-trapped, to the phonon energy. Thus, the weak-coupling regime, characterized by $\alpha < 1$, indicates that the long-range electron–phonon coupling is too weak to produce self-trapping. Expressed in terms of α, the zero-temperature weak-coupling polaron energy associated with the long-range electron–phonon coupling given by Eq. (3.1) is simply

$$\Delta E_{\mathrm{WC}}^{\mathrm{LR}} = -\alpha\hbar\omega. \tag{3.4}$$

The corresponding effective mass for a weak-coupling polaron is also readily described in terms of α:

$$\frac{m^*}{m} = 1 + \frac{\alpha}{6}. \tag{3.5}$$

Since $\alpha < 1$ it is evident that the effective mass of a weak-coupling polaron is only slightly higher than that of the bare carrier.

3.2 Short-range electron–phonon interactions

The short-range electron–phonon interaction also lowers the energy of a carrier at the bottom of a conduction band. However, the short range of the electron–phonon interaction fosters a more effective interaction with short-wavelength phonons than occurs with the long-range electron–phonon interaction. Treating the short-range

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electron–phonon interaction as a perturbation gives the corresponding zero-temperature weak-coupling polaron energy:

$$\Delta E_{\mathrm{WC}}^{\mathrm{SR}} = -\frac{F^2/k}{12\left(R_{\mathrm{p}}/a\right)^2}. \tag{3.6}$$

The difference in the ranges of their electron–phonon interactions accounts for the different dependences of $\Delta E_{\mathrm{WC}}^{\mathrm{LR}}$ and $\Delta E_{\mathrm{WC}}^{\mathrm{SR}}$ on R_{p} that are manifested between Eq. (3.1) and Eq. (3.6).

The condition governing breakdown of a short-range electron–phonon interaction's weak-coupling solution yields a condition that can be understood with a classical argument (Emin, 1973a). In particular, the weak-coupling solution fails if the product of the force that a carrier exerts on adjacent atoms multiplied by the time that a carrier can spend at a site exceeds the momentum of these atoms' vibrations: $F(\hbar/t) > P_{\mathrm{vib}}$. Atoms' vibratory momentum increases with rising temperature, thereby increasing the stability of the weak-coupling solution. At zero temperature, with only zero-point vibrations, the weak-coupling solution fails when $[(F^2/k)\hbar\omega]^{1/2} > t$.

3.3 Two-site model: origin of carrier-induced softening

A general feature of the weak-coupling polaron is that its energy vanishes in the limit that atoms' vibration frequencies vanish. In particular, $\Delta E_{\mathrm{WC}}^{\mathrm{LR}}$ and $\Delta E_{\mathrm{WC}}^{\mathrm{SR}}$ of Eq. (3.1) and Eq. (3.6) vanish since R_{p} tends to infinity with diminishing atomic-vibration frequency. By contrast, the energy lowering associated with self-trapping, discussed in Chapter 4, will be seen to remain finite in this limit. This distinction indicates a fundamental difference in the physics underlying weak-coupling and strong-coupling polarons.

To elucidate this difference consider the model of Fig. 3.1 in which an electron transfers between two independent equivalent harmonically vibrating molecules. Deformation of each of the two molecules is characterized by a single distortion parameter, Δ_i with $i = 1$ or 2. The Hamiltonian describing carrier-free harmonic vibrations of these molecules is:

$$H_{\mathrm{vib}} = -\frac{\hbar^2}{2M}\left(\frac{\partial^2}{\partial\Delta_1^2} + \frac{\partial^2}{\partial\Delta_2^2}\right) + \frac{k}{2}\left(\Delta_1^2 + \Delta_2^2\right), \tag{3.7}$$

where M is the reduced mass associated with a molecule's vibrations and k is the associated spring constant. A short-range electron–phonon interaction results from

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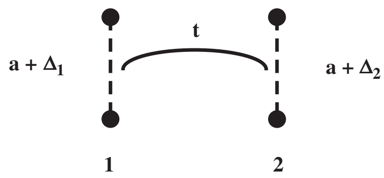

Fig. 3.1 An electron transfers with energy t between two molecules whose deformation parameters are Δ_1 and Δ_2, respectively.

the energy of the electron occupying a molecule depending upon the deformation of that molecule. With the electron occupying the i-th molecule:

$$\varepsilon_{\mathrm{SR},i} = -F\Delta_i. \tag{3.8}$$

The electron's intermolecular motion is associated with the transfer energy t.

In the spirit of the tight-binding approach, an electronic eigenstate is expressed as a superposition of local states associated with its occupation of each of the two molecules:

$$\Psi = a_1\Phi_1 + a_2\Phi_2. \tag{3.9}$$

The expansion coefficients are found by solving the eigenvalue matrix equation:

$$\begin{bmatrix} -F\Delta_1 - W & t \\ t & -F\Delta_2 - W \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0. \tag{3.10}$$

The roots of the associated determinant are the two electronic eigenvalues:

$$W_{\pm} = -\frac{F(\Delta_2 + \Delta_1)}{2} \pm \sqrt{\left[\frac{F(\Delta_2 - \Delta_1)}{2}\right]^2 + t^2}. \tag{3.11}$$

In the absence of intermolecular transfer, $t = 0$, the eigenvalues are just the electronic energies for the electron confined on one or the other of the two molecules: $W = -F\Delta_1$ and $W = -F\Delta_2$. In the complementary limit of vanishing electron–phonon interaction, $F = 0$, the eigenvalues are t and $-t$, the upper and lower states of the two-state energy band associated with intermolecular transfer.

The molecule occupied by the electron is governed by the relative values of their deformation parameters. It is therefore expeditious to introduce the relative deformation parameter $x \equiv (\Delta_2 - \Delta_1)/\sqrt{2}$ along with the complementary parameter

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$X \equiv (\Delta_2 + \Delta_1)/\sqrt{2}$. Rewriting the vibration Hamiltonian of Eq. (3.7) and the electronic eigenvalues of Eq. (3.11) in terms of these parameters yields:

$$H_{\text{vib}} = -\frac{\hbar^2}{2M}\left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial x}\right) + \frac{k}{2}\left(X^2 + x^2\right) \tag{3.12}$$

and

$$W_{\pm} = -\frac{FX}{\sqrt{2}} \pm \sqrt{\frac{(Fx)^2}{2} + t^2}. \tag{3.13}$$

The x-dependent portion of the sum of the vibration and electronic potentials governs the electron's intermolecular motion:

$$V_{\pm}(x) \equiv \frac{k}{2}x^2 \pm \sqrt{\frac{(Fx)^2}{2} + t^2}. \tag{3.14}$$

As illustrated in Fig. 3.2, increasing the strength of the electron–phonon interaction softens the potential of the lower electronic state $V_-(x)$ and stiffens the potential of the upper electronic state $V_+(x)$ (Öpik and Pryce, 1957).

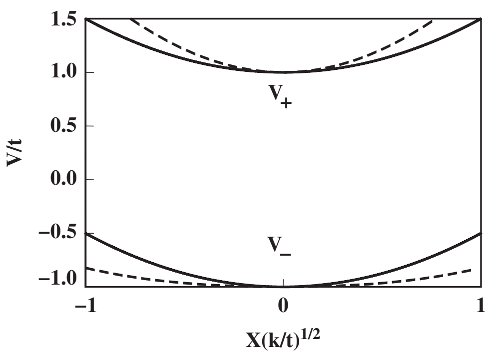

Fig. 3.2 The sum of the molecular-vibration and electronic energies (in units of the electronic transfer energy t) is plotted against the relative deformation parameter x (in units of $\sqrt{t/k}$) for the two electronic branches. Increasing the electron–phonon coupling strength from zero (solid curves) to a finite value (dashed curves) reduces the curvature of the lower electronic branch, V_-, and increases the curvature of the upper electronic branch, V_+.

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Both potentials remain harmonic for sufficiently weak electron–phonon coupling:

$$V_{\pm}(x) \cong \pm t + \frac{[k \pm (F^2/2t)]}{2} x^2. \tag{3.15}$$

Softening of the harmonic ground-state potential reduces its zero-point energy by

$$\begin{aligned} \Delta E &= \frac{\hbar}{2}\left(\sqrt{\frac{k-F^2/2t}{M}}-\sqrt{\frac{k}{M}}\right) \\ &= \frac{\hbar\omega}{2}\left(\sqrt{1-F^2/2kt}-1\right) \approx -\frac{F^2/k}{8(R_p/a)^2}, \end{aligned} \tag{3.16}$$

where R_p is defined in Eq. (3.2) and the numerical coefficient for this two-site model differs by a factor of 3/2 from that of Eq. (3.6) obtained for a three-dimensional cubic lattice. In a similar manner, stiffening of the potential for the upper state increases its zero-point energy by $|\Delta E|$. The energy separation between the lower and upper state is thereby increased. More generally, as depicted in Fig. 3.3, the electron–phonon interaction produces a temperature increase of the width of an energy band comprising weak-coupling states.

The processes that produce this softening and stiffening can be understood by considering how the electron moves in response to atomic vibrations. Solution of Eq. (3.10) describes the transfer of the electron upon changing the relative molecular deformation parameter:

$$P_{\pm}(y) \equiv |a_2^{\pm}|^2 - |a_1^{\pm}|^2 = \frac{\left(-y \pm \sqrt{1+y^2}\right)^2 - 1}{\left(-y \pm \sqrt{1+y^2}\right)^2 + 1} \cong \mp y, \tag{3.17}$$

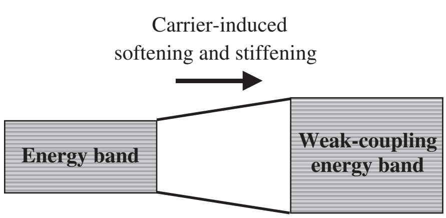

Fig. 3.3 Electron–phonon coupling lowers the interacting system's energy for low-energy electronic states and raises the interacting system's energy for high-energy electronic states. The electron–phonon interaction thereby broadens the band of system energies.

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where $y \equiv Fx/\sqrt{2}t = F(\Delta_2 - \Delta_1)/2t = (\varepsilon_{SR,1} - \varepsilon_{SR,2})/2t$. As illustrated in Fig. 3.4 for the (negative) root associated with softening, the electron moves from molecule 1 to molecule 2 as its electronic energy falls below that of molecule 1. By contrast, for the (positive) root associated with stiffening, the electron moves from molecule 2 to molecule 1 despite its electronic energy rising above that of molecule 2. That is, softening occurs as the electron moves toward the lower energy site thereby enhancing its molecule's deformation. However, stiffening occurs when the electron moves away from the lower energy site thereby reducing its molecule's deformation.

The harmonic approximation of Eq. (3.15) is valid if the amplitude of a zero-point vibration is less than the maximum value of x for which the potential of Eq. (3.14) remains quadratic: $\sqrt{\hbar\omega/k} < \sqrt{2}t/F$. This condition, $\sqrt{(F^2/2k)\hbar\omega} <$ t, is essentially just that for the validity of the zero-temperature weak-coupling polaron in a cubic crystal (Emin, 1972; Emin 1973a). In essence, the weak-coupling solution describes a carrier whose adjustment to displacements of atoms just modifies the stiffness constants that govern their vibrations.

A different type of solution, corresponding to a self-trapped carrier, emerges when $F^2/2k > t$. In this circumstance, as shown in Fig. 3.5, $V_{-}(x)$ of Eq. (3.14) garners two minima (Öpik and Pryce, 1957). As $F^2/2kt$ is increased the two minima move from $x \approx 0$ toward $x = \pm F/\sqrt{2}k$. Deforming the molecules to establish one of these minima corresponds to localizing the electron on one of the two molecules while lowering its ground-state energy by $F^2/2k$.

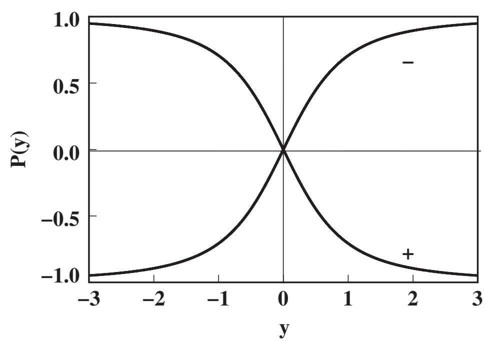

Fig. 3.4 The probability of the electron occupying molecule 2 versus occupying molecule 1 is plotted for positive and negative branches of the electronic spectrum. $P(y) > 0$ indicates that molecule 2 is preferentially occupied and $P(y) < 0$ indicates that molecule 1 is preferentially occupied. Occupation of molecule 1 or 2 is energetically preferred for $y < 0$ and $y > 0$, respectively. Thus the negative branch has the electron occupying the energetically preferred molecule. However, the positive branch has the carrier occupying the energetically unfavorable molecule.

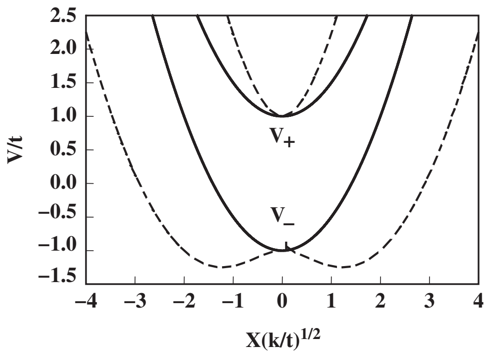

Fig. 3.5 The sum of the molecular-vibration and electronic energies (in units of the electronic transfer energy t) is plotted against the relative deformation parameter x (in units of $\sqrt{t/k}$) for the two electronic branches. Increasing the electron–phonon coupling strength from zero (solid curves) to a large enough value (dashed curves) increases the curvature of the upper electronic branch, V_+, while producing two symmetrically placed self-trapping minima on the lower electronic branch, V_-.

This section illustrated carrier-induced softening by considering a charge carrier that can move between two sites. However, this example does not correctly depict self-trapping. Self-trapping in a solid cannot be adequately modeled with just two sites. The following chapters will address self-trapping in considerably more detail.

The adiabatic limit reveals a qualitative difference between a weak-coupling polaron and a strong-coupling polaron. The adiabatic limit envisions letting atomic masses approach infinity while keeping interatomic stiffness constants finite. Atomic vibration frequencies vanish in the adiabatic limit. The energy lowering of a weak-coupling polaron results because a carrier's interactions with atoms alter their vibration frequencies. Therefore a weak-coupling polaron's energy lowering vanishes in the adiabatic limit. By contrast, a strong-coupling polaron is based on the self-trapping that occurs as a carrier's interactions with atoms shift their equilibrium positions. These shifts depend on the stiffness constants governing the associated atomic displacements. The energy lowering of a strong-coupling polaron remains finite in the adiabatic limit.

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